

INTEGER QUANTIZATION FOR DEEP LEARNING INFERENCE: PRINCIPLES AND EMPIRICAL EVALUATION

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ABSTRACT

Quantization techniques can reduce the size of Deep Neural Networks and improve inference latency and throughput by taking advantage of high throughput integer instructions. In this paper we review the mathematical aspects of quantization parameters and evaluate their choices on a wide range of neural network models for different application domains, including vision, speech, and language. We focus on quantization techniques that are amenable to acceleration by processors with high-throughput integer math pipelines. We also present a workflow for 8-bit quantization that is able to maintain accuracy within 1% of the floating-point baseline on all networks studied, including models that are more difficult to quantize, such as MobileNets and BERT-large.

1 Introduction

While 32-bit single-precision floating-point was the dominant numerical format for Deep Learning (DL) applications, more recently a variety of alternative formats have been proposed to increase the computational performance of deep learning applications. It is becoming commonplace to train neural networks in 16-bit floating-point formats, either IEEE fp16 [35] or bfloat16 [57], supported by most DL accelerators. Once trained, neural networks can be deployed for inference using even lower-precision formats, including floating-point, fixed-point, and integer. Low-precision formats offer several performance benefits. First, many processors provide higher throughput math pipelines the low-bit formats, which can speed up math-intensive operations, such as convolutions and matrix multiplications. Second, smaller word sizes reduce memory bandwidth pressure, improving performance for bandwidth-limited computations. Third, smaller word sizes lead to lower memory size requirements, which can improve cache utilization as well as other aspects of memory-system operation.

In this paper we focus on integer quantization for neural network inference, where trained networks are modified to use integer weights and activations so that integer math pipelines can be used for many operations. Table 1 lists the relative tensor operation throughputs of various data types on the NVIDIA Turing Graphics Processing Unit (GPU) architecture [40]. Math-intensive tensor operations executed on 8-bit integer types can see up to a 16x speed-up compared to the same operations in fp32. Memory-limited operations could see up to a 4x speed-up compared to the fp32 version, due to the smaller word size. Other processors, such as TPUv1 [23], Intel CPUs with VNNI

Input Data type	Accumulation Data type	Math Throughput	Bandwidth Reduction
FP32	FP32	1x	1x
FP16	FP16	8x	2x
INT8	INT32	16x	4x
INT4	INT32	32x	8x
INT1	INT32	128x	32x

Table 1: Benefits of lower precision data types for tensor operations on the NVIDIA Turing GPU architecture

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